# Automatic extraction of semantic AOIs from heterogeneous SVG visualizations

***Project Plan***

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*Under the supervision of Anne-Flore Cabouat.*

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## Project Overview

### Description

This project's goal is to extract and visualize the *Area of Interests* (**AOI**s) of heterogeneous SVG files (Scalable Vector Graphics). The SVGs we are working on are already data visualizations, and have a somewhat shared structure (legends, labels, axes, ETC.). This extraction's goal is to better interpret eye-tracking data. Eye-tracking methods record where users look on a chart or map, but only as coordinates on the screen. Without additional information, these coordinates $(x, y)$ are hard to interpret, since we don't know what part of the visualization the user is actually looking at, we only know their placement on the screen.

To make this data meaningful, it needs to be associated with specific elements of a visualization, such as data points, axes or legend items. This way, instead of just having positions, we can understand how users interact with the visualization.

However the challenge is that SVG files are not organized in a consistent way. Different types of visualizations (maps, treemaps, scatterplots) use different structures, which makes it challenging to exract a unified representation. Additionally, important elements are not always clearly labeled or explicitly defined in SVG files.

Our goal is therefore to build a pipeline that converts heterogeneous SVG visualizations into a standardized AOI representation, which can be later be used for eye-tracking data analysis.

### Tasks & Approach

To achieve this, our pipeline must include the following steps:

- Parse SVG files,
- Identify meaningful elements such as data marks, axes, labels, and legend items,
- Retrieve semantic information from IDs, classes and group hierarchy,
- Compute bounding boxes for each element,
- Construct a standardized AOI table,
- Build a hierarchical tree representation of each visualization,
- Validate extracted AOIs through manual inspection and consistency checks.

### Data

This project relies on 2 main data sources:

- **Structured SVG files (16 visualizations)** representing heterogeneous visualization types (maps, treemaps, scatterplots, ETC.). These will be the input for AOI extraction.
- **Pilot eye-tracking data** containing recorded user gaze coordinates. This data will later be used to validate the extracted AOIs and study how users look at different parts of a visualization.

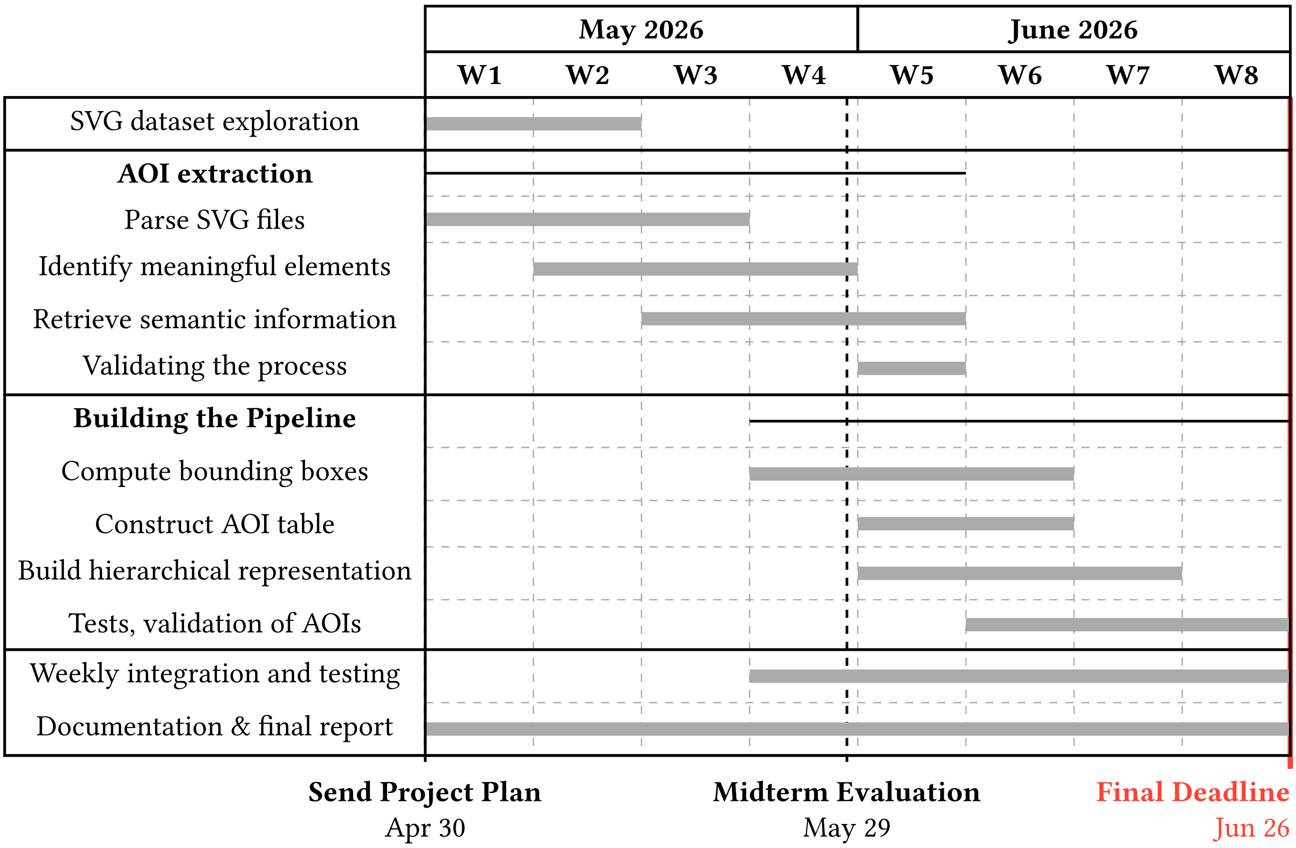
## Methods

To turn our project into reality, we'll apply the following methods:

- **Hierarchical SVG parsing**: treating each visualization as a structured tree that can be traversed and analyzed.
- **Rule-based semantic extraction**: using SVG tags, IDs, classes and grouping structure to identify meaningful visual elements.
- **Geometric AOI modeling**: using bounding boxes and spatial representations to define AOIs.
- **Normalization procedures**: to convert heterogeneous SVG structures into a standardized AOI representation.

# Timeline

Below is an indicative yet realistic schedule developed by the team. We have aimed for a balance between being specific and not overly exhaustive, ensuring this schedule provides a high-level overview of our project's progress.




# Roles & Member Tasks

The following is an initial breakdown of the roles we have assigned ourselves for the project. The goal of this allocation is to ensure clear ownership and accountability for every aspect of the project. However, these roles remain flexible and may be adjusted as the project evolves.

## Shared Responsabilities

- Development of the AOI extraction pipeline
- Weekly integration and testing
- Validation and refinement of extraction rules
- Continuous documentation
- Midterm and final presentations

### SVG Exploration (by visualization types)

- **C. ABOU KHEIR:** ***Mixed Visualization Lead***
  - Explore edge cases and less structured visualizations.
  - Identify potential challenges for normalization.
- **J. P. AL AM:** ***Cross-Visualization and Integration Lead***
  - Compare structural patterns accross all visualizations types.
  - Support development of common parsing and normalization rules.
  - Coordinate integration of findings from all members.
- **A. MALLET:** ***Scatterplot Lead***
  - Analyze point marks, axes, labels and legend structures.
  - Document reccuring semantic patterns.
- **M-J. TANNOURY:** ***Maps Lead***
  - Analyze SVG structure and semantic patterns specific to maps.
  - Identify challenges related to geographic marks and labels.
- **A. TRAD:** ***Treemap Lead***
  - Explore rectangle hierarchies and grouping structures.
  - Identify AOI extraction challenges specific to treemaps.

### Functional lead roles during implementation

- **C. ABOU KHEIR:** ***Validation support***
- **J. P. AL AM:** ***Integration and documentation support***
- **A. MALLET:** ***Geometry/AOI modeling support***
- **M-J. TANNOURY:** ***Parsing support***
- **A. TRAD:** ***Semantic extraction support***

## Risks & Mitigation plans

| Risks | Mitigation Plans |
|---|---|
| SVG structures vary significantly accross visualizations, which makes it difficult to apply a single extraction approach. | Begin with a common AOI schema and ajust parsing rules iteratively. |
| Important semantic information may be missing or inconsistently labeled in some SVG files. | Use the position of the elements and their grouping in the SVG to better identify AOIs whose labels are missing or unclear. |
| Bounding boxes or AOIs may be innacurate for complex visual elements. | Perform manual validation and consistency checks on sample visualizations. |
| Integration challenges and issues may occur when combining parsing, geometry and normalization. | Use shared code repository and perform weekly integration. |
| Project timeline may be underestimated due to unforeseen complexity. | Prioritize completing a core AOI extraction pipeline first, leaving optional ajustements for later. |